

Development Roadmap for Pharmacy Management System

Desktop Application Offline Software

(Version 1.0)

Stage 1 – Project Foundation

Goal: Build a stable foundation for the entire software.

Phase 1.1 – Project Initialization

- Create project structure
 - Configure frontend
 - Configure backend
 - Configure desktop application
 - Environment configuration
 - Git initialization
 - Run first successful application
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Phase 1.2 – Database Foundation

- SQLite setup
 - Database connection
 - ORM configuration
 - Migration system
 - Database initialization
 - Seed mechanism
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Phase 1.3 – Application Core

- Global configuration

- Logging system
 - Error handling
 - Exception handling
 - Validation framework
 - Constants
 - Utility functions
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Phase 1.4 – UI Foundation

- Theme
 - Color system
 - Typography
 - Icons
 - Layout
 - Sidebar
 - Header
 - Footer
 - Routing
 - Responsive desktop layout
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Phase 1.5 – Shared Components

- Buttons
- Inputs
- Cards
- Dialogs
- Tables
- Pagination

- Search
 - Toasts
 - Loading
 - Empty states
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Stage 2 – Login & License Management

Goal: Secure the software.

Phase 2.1 – Hardware ID System

- Generate Hardware ID
 - Display Hardware ID
 - Copy Hardware ID
 - Machine fingerprint
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Phase 2.2 – License Engine

- License file
 - License encryption
 - License validation
 - Expiry calculation
 - Lifetime license
 - Invalid license detection
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Phase 2.3 – License Activation UI

- Activation screen
- Import license
- Activate
- License information

- Expired license screen
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Phase 2.4 – Authentication

- Login
 - Logout
 - Password hashing
 - Session
 - Remember me
 - Change password
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Phase 2.5 – Startup Flow

- Splash screen
 - License verification
 - Login redirection
 - Session restore
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Stage 3 – Masters

Goal: Create all master/reference data.

Phase 3.1 – Categories

- CRUD
 - Search
 - Status
-

Phase 3.2 – Companies

- CRUD
- Search

- Status
-

Phase 3.3 – Suppliers

- CRUD
 - Contact information
 - Search
-

Phase 3.4 – Customers

- CRUD
 - Search
 - Balance
-

Phase 3.5 – Medicines

- Complete medicine form
 - Barcode
 - Purchase price
 - Sale price
 - Expiry
 - Stock settings
 - Company
 - Category
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Phase 3.6 – Medicine Utilities

- Barcode generation
- Medicine search
- Quick filters

- Import/export medicines
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Stage 4 – Purchase Management

Goal: Bring stock into the pharmacy.

Phase 4.1 – Purchase Invoice

- New purchase
 - Supplier selection
 - Invoice information
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Phase 4.2 – Purchase Items

- Medicine selection
 - Quantity
 - Cost
 - Discount
 - Tax
 - Totals
-

Phase 4.3 – Purchase Processing

- Save purchase
 - Update stock
 - Purchase history
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Phase 4.4 – Purchase Return

- Return purchase
- Stock deduction
- History

Phase 4.5 – Purchase Printing

- Invoice preview
- Print
- PDF

Stage 5 – Inventory Management

Goal: Maintain accurate stock.

Phase 5.1 – Stock Management

- Current stock
- Stock value
- Stock summary

Phase 5.2 – Stock Adjustment

- Increase
- Decrease
- Adjustment reason

Phase 5.3 – Stock Monitoring

- Low stock
- Out of stock
- Overstock

Phase 5.4 – Expiry Management

- Expiring medicines
- Expired medicines

- Alerts
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Phase 5.5 – Inventory History

- Stock movement
 - Adjustment history
 - Audit
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Stage 6 – Sales & POS Billing

Goal: Sell medicines efficiently.

Phase 6.1 – POS Layout

- POS screen
 - Search
 - Barcode
 - Cart
-

Phase 6.2 – Billing

- Quantity
 - Discount
 - Tax
 - Total
-

Phase 6.3 – Payment

- Cash
 - Change
 - Complete sale
-

Phase 6.4 – Invoice

- Receipt
 - Print
 - PDF
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Phase 6.5 – Sales Return

- Return invoice
 - Stock update
 - Refund
-

Phase 6.6 – Sales History

- View invoices
 - Search
 - Reprint
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Stage 7 – Dashboard

Goal: Show business insights.

Phase 7.1 – Summary Cards

Daily

- Today's Sales
- Today's Purchases
- Today's Profit
- Today's Transactions

Masters

- Total Medicines
- Total Customers
- Total Suppliers

Inventory

- Current Stock Value
- Out of Stock
- Low Stock
- Expiring Soon
- Expired Medicines

Financial

- Total Sales
 - Total Purchases
 - Net Profit
-

Phase 7.2 – Charts

- Sales chart
 - Purchase chart
 - Profit chart
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Phase 7.3 – Activity Widgets

- Recent sales
 - Low stock
 - Expiry alerts
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Phase 7.4 – Dashboard Filters

- Today
 - Week
 - Month
 - Year
 - Custom
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Stage 8 – Reports

Goal: Analyze all business data.

Phase 8.1 – Sales Reports

- Daily
 - Monthly
 - Yearly
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Phase 8.2 – Purchase Reports

- Supplier
 - Date
 - Invoice
-

Phase 8.3 – Inventory Reports

- Stock
 - Movement
 - Valuation
-

Phase 8.4 – Medicine Reports

- Expiry
 - Low stock
 - Fast moving
 - Slow moving
-

Phase 8.5 – Financial Reports

- Profit
- Expenses (if implemented)

- Revenue
-

Phase 8.6 – Export & Print

- PDF
 - Excel
 - Print
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Stage 9 – Backup & Restore

Goal: Protect customer data.

Phase 9.1 – Manual Backup

- Create backup
 - Backup location
 - Backup naming
 - Compression
 - Progress
 - Success message
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Phase 9.2 – Restore

- Browse backup
- Validate backup
- Create safety backup

- Restore database
 - Restart application
 - Result message
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Phase 9.3 – Automatic Backup

- Schedule
 - Retention
 - Auto delete
 - Startup backup
 - Notifications
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Phase 9.4 – Backup History

- View backups
 - Search
 - Restore
 - Open folder
 - Delete
 - Verify
-

Phase 9.5 – Backup Verification

- Integrity check
 - Corruption detection
 - Database validation
 - Verification status
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Phase 9.6 – Backup Settings

- Backup folder
 - Automatic backup
 - Schedule
 - Retention
 - Compression
 - Verification
 - Startup backup
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Stage 10 – Settings

Goal: Configure the software.

Stage 10 – Settings

Phase 10.1 – Pharmacy Profile

- Pharmacy Information
 - Contact Information
 - Logo
 - Registration Information
 - Receipt Identity
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Phase 10.2 – Billing & POS Settings

- Currency
 - Tax
 - Discount
 - Invoice Numbering
 - Payment Defaults
 - POS Behavior
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Phase 10.3 – Inventory & Medicine Settings

- Low Stock Threshold
 - Expiry Alert
 - Negative Stock
 - Default Units
 - Barcode Settings
 - Expired Medicine Control
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Phase 10.4 – Printer & Receipt Settings

- Printer Selection
 - Thermal Printer
 - A4 Printer
 - Paper Size
 - Receipt Customization
 - Receipt Preview
 - Test Print
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Phase 10.5 – Appearance & System Preferences

- Theme
 - Date Format
 - Time Format
 - Number Format
 - Startup Behaviour
 - Notifications
 - Language Future-Ready
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Phase 10.6 – Backup & Data Management

- Backup Preferences
- Backup Location
- Automatic Backup
- Retention
- Database Information

- Database Health
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Phase 10.7 – Security & Maintenance

- Change Password
 - Auto Lock
 - Session Timeout
 - Database Optimization
 - Integrity Check
 - Application Logs
 - Clear Temporary Data
 - Safe Data Reset
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Phase 10.8 – License & About

- License Status
 - License Type
 - Activation Date
 - Expiry
 - Remaining Days
 - Hardware ID
 - License Renewal/Import
 - Software Version
 - Build Number
 - Developer Information
 - System Information
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